

For this mechanism, I will use the 1.5 mm module because it gives larger and more printable teeth than module 1 mm.

The calculation below uses current design:

Rack load = 50 N upward

5 kg payload

Arm length = 300 mm

Pinion outside diameter = 30 mm

Pressure angle = 20°

Material = PLA

Factor of safety = 2.5

Required rack travel = 30 to 50 mm

The mechanism is treated as:

Rack $\rightarrow$ Spur gear $\rightarrow$ Sector gear + 300 mm arm $\rightarrow$ 5 kg payload

The pinion controls the rack motion and rotates about its own axis.

1. Basic design parameters

Parameter	Symbol	Selected value
Module	m	1.5 mm
Pressure angle	ϕ	20°
Pinion teeth	Z_p	18
Spur teeth	Z_{sp}	30
Sector teeth	Z_s	66
Pinion outside diameter	d_{op}	30 mm
Spur outside diameter	d_{osp}	48 mm
Sector outside diameter	d_{os}	102 mm
Arm length	L_a	300 mm
Payload	M	5 kg
Direct rack force	F_D	50 N

Parameter	Symbol	Selected value
Safety factor	SF	2.5

2. Pinion tooth calculation

For a standard spur gear:

$$d = mZ$$

Outside diameter:

$$d_o = m(Z + 2)$$

You specified:

$$d_o = 30 \text{ mm}$$

Therefore:

$$30 = 1.5(Z + 2)$$

$$Z + 2 = 20$$

$$Z_p = 18 \text{ teeth}$$

Pitch diameter:

$$d_p = mZ_p$$

$$d_p = 1.5(18)$$

$$d_p = 27 \text{ mm}$$

Pitch radius:

$$r_p = \frac{27}{2}$$

$$r_p = 13.5 \text{ mm}$$

3. Pinion tooth profile

For a standard 20° full-depth involute gear:

Addendum:

$$h_a = m$$

$$h_a = 1.5 \text{ mm}$$

Dedendum:

$$h_f = 1.25m$$

$$h_f = 1.875 \text{ mm}$$

Whole tooth depth:

$$h = 2.25m$$

$$h = 3.375 \text{ mm}$$

Outside diameter:

$$d_o = 30 \text{ mm}$$

Root diameter:

$$d_f = d - 2.5m$$

$$d_f = 27 - 2.5(1.5)$$

$$d_f = 23.25 \text{ mm}$$

Base diameter:

$$d_b = d \cos \phi$$

$$d_b = 27 \cos 20^\circ$$

$$d_b = 25.37 \text{ mm}$$

Circular pitch:

$$p = \pi m$$

$$p = \pi(1.5)$$

$$p = 4.712 \text{ mm}$$

Tooth thickness at pitch circle:

$$t = \frac{p}{2}$$

$$t = 2.356 \text{ mm}$$

4. Spur gear calculation

Select:

$$\boxed{Z_{sp} = 30}$$

Pitch diameter:

$$\begin{aligned}d_{sp} &= mZ_{sp} \\d_{sp} &= 1.5(30) \\ \boxed{d_{sp} &= 45 \text{ mm}}\end{aligned}$$

Outside diameter:

$$\begin{aligned}d_{o,sp} &= m(Z_{sp} + 2) \\d_{o,sp} &= 1.5(32) \\ \boxed{d_{o,sp} &= 48 \text{ mm}}\end{aligned}$$

Root diameter:

$$\begin{aligned}d_{f,sp} &= m(Z_{sp} - 2.5) \\d_{f,sp} &= 1.5(27.5) \\ \boxed{d_{f,sp} &= 41.25 \text{ mm}}\end{aligned}$$

Base diameter:

$$\begin{aligned}d_{b,sp} &= 45 \cos 20^\circ \\ \boxed{d_{b,sp} &= 42.29 \text{ mm}}\end{aligned}$$

Pitch radius:

$$\boxed{r_{sp} = 22.5 \text{ mm}}$$

5. Pinion to spur center distance

For two external gears:

$$a = \frac{d_p + d_{sp}}{2}$$
$$a = \frac{27 + 45}{2}$$
$$\boxed{a = 36 \text{ mm}}$$

Therefore, the pinion and spur gear centers must be:

$$\boxed{36 \text{ mm apart}}$$

6. Sector gear calculation

The sector gear needs to generate the torque required to lift the 5 kg payload.

Payload force:

$$W = Mg$$
$$W = 5(9.81)$$
$$\boxed{W = 49.05 \text{ N}}$$

Arm length:

$$L_a = 300 \text{ mm} = 0.3 \text{ m}$$

Required sector torque:

$$T_s = WL_a$$
$$T_s = 49.05(0.3)$$
$$\boxed{T_s = 14.715 \text{ N} \cdot \text{m}}$$

7. Selecting sector gear teeth

Recommended:

$$\boxed{Z_s = 66}$$

Pitch diameter:

$$d_s = mZ_s$$
$$d_s = 1.5(66)$$
$$\boxed{d_s = 99 \text{ mm}}$$

Pitch radius:

$$r_s = \frac{99}{2}$$
$$\boxed{r_s = 49.5 \text{ mm}}$$

Outside diameter:

$$d_{o,s} = m(Z_s + 2)$$
$$d_{o,s} = 1.5(68)$$
$$\boxed{d_{o,s} = 102 \text{ mm}}$$

Root diameter:

$$d_{f,s} = m(Z_s - 2.5)$$
$$d_{f,s} = 1.5(63.5)$$
$$\boxed{d_{f,s} = 95.25 \text{ mm}}$$

Base diameter:

$$d_{b,s} = 99 \cos 20^\circ$$
$$\boxed{d_{b,s} = 93.08 \text{ mm}}$$

8. Force required at sector gear

The tangential force at the sector pitch circle is:

$$F_s = \frac{T_s}{r_s}$$
$$F_s = \frac{14.715}{0.0495}$$
$$\boxed{F_s = 297.27 \text{ N}}$$

This is the approximately 300 N force you mentioned.

Therefore:

$$\boxed{F_s \approx 297 \text{ N}}$$

9. Sector force with safety factor

Using:

$$SF = 2.5$$

$$F_{s,design} = F_s(SF)$$

$$F_{s,design} = 297.27(2.5)$$

$$\boxed{F_{s,design} = 743.18 \text{ N}}$$

The sector-spur tooth mesh should therefore be designed to withstand approximately:

$$\boxed{743 \text{ N}}$$

10. Rack force

You have two opposing forces on the rack.

Upward direct force:

$$F_D = 50 \text{ N}$$

Downward force transmitted from the sector through the spur:

$$F_I = 297.27 \text{ N}$$

Therefore:

$$F_R = F_I - F_D$$

$$F_R = 297.27 - 50$$

$$\boxed{F_R = 247.27 \text{ N downward}}$$

This is the net rack force.

11. Rack design force with FOS

$$F_{R,design} = F_R(SF)$$

$$F_{R,design} = 247.27(2.5)$$

$$\boxed{F_{R,design} = 618.18 \text{ N}}$$

Therefore, the pinion-rack system should be designed for approximately:

$$\boxed{618 \text{ N}}$$

net design load.

Important distinction:

The rack-spur mesh carries approximately:

$$\boxed{297.27 \text{ N actual}}$$

or:

$$\boxed{743.18 \text{ N design}}$$

The pinion sees the net rack force:

$$\boxed{247.27 \text{ N actual}}$$

or:

$$\boxed{618.18 \text{ N design}}$$

12. Spur gear torque

The spur gear pitch radius is:

$$r_{sp} = 22.5 \text{ mm} = 0.0225 \text{ m}$$

Therefore:

$$\begin{aligned} T_{sp} &= F_s r_{sp} \\ T_{sp} &= 297.27(0.0225) \\ \boxed{T_{sp} &= 6.688 \text{ N} \cdot \text{m}} \end{aligned}$$

With FOS:

$$\begin{aligned} T_{sp,design} &= 6.688(2.5) \\ \boxed{T_{sp,design} &= 16.72 \text{ N} \cdot \text{m}} \end{aligned}$$

13. Check of spur to sector gear ratio

For external gears:

$$\frac{T_s}{T_{sp}} = \frac{Z_s}{Z_{sp}}$$

Therefore:

$$\begin{aligned} T_s &= T_{sp} \frac{Z_s}{Z_{sp}} \\ T_s &= 6.688 \left(\frac{66}{30} \right) \end{aligned}$$

$$T_s = 14.715 \text{ N} \cdot \text{m}$$

So the 30-tooth spur and 66-tooth sector combination produces exactly the required theoretical sector torque.

14. Spur to sector center distance

$$a_{sp-s} = \frac{d_{sp} + d_s}{2}$$

$$a_{sp-s} = \frac{45 + 99}{2}$$

$$a_{sp-s} = 72 \text{ mm}$$

15. Pinion torque

The pinion experiences the net rack force:

$$F_R = 247.27 \text{ N}$$

Pinion pitch radius:

$$r_p = 13.5 \text{ mm} = 0.0135 \text{ m}$$

Therefore:

$$T_p = F_R r_p$$

$$T_p = 247.27(0.0135)$$

$$T_p = 3.338 \text{ N} \cdot \text{m}$$

With FOS:

$$T_{p,design} = 3.338(2.5)$$

$$T_{p,design} = 8.35 \text{ N} \cdot \text{m}$$

16. Rack dimensions

Rack module must match the gears:

$$m = 1.5 \text{ mm}$$

Rack pitch:

$$p = \pi m$$

$$p = 4.712 \text{ mm}$$

Rack tooth thickness:

$$t = \frac{p}{2}$$

$$t = 2.356 \text{ mm}$$

Rack addendum:

$$h_a = 1.5 \text{ mm}$$

Rack dedendum:

$$h_f = 1.875 \text{ mm}$$

Total tooth depth:

$$h = 3.375 \text{ mm}$$

17. Number of rack teeth required

For maximum required travel of 50 mm:

$$N_R = \frac{L}{p}$$

$$N_R = \frac{50}{4.712}$$

$$N_R = 10.61$$

You cannot use 10.61 teeth.

Use at least:

$$12 \text{ teeth}$$

Rack tooth length:

$$L_R = 12p$$

$$L_R = 12(4.712)$$

$$L_R = 56.55 \text{ mm}$$

This provides:

$$56.55 \text{ mm}$$

of theoretical tooth engagement length.

I recommend making the physical rack longer than the tooth length to provide mounting and end clearance.

A practical prototype rack can therefore be approximately:

$$\boxed{75 \text{ to } 80 \text{ mm long}}$$

with 12 teeth.

18. Pinion rotation for rack movement

The fundamental relationship is:

$$L = \frac{\theta}{360^\circ} \pi d_p$$

Therefore:

$$\theta = \frac{360^\circ L}{\pi d_p}$$

For 30 mm:

$$\theta = \frac{360(30)}{\pi(27)}$$
$$\boxed{\theta = 127.3^\circ}$$

For 50 mm:

$$\theta = \frac{360(50)}{\pi(27)}$$
$$\boxed{\theta = 212.2^\circ}$$

Therefore:

$$\boxed{127.3^\circ \leq \theta \leq 212.2^\circ}$$

19. Pinion rotation direction

If the pinion rotates clockwise:

$$\boxed{\text{Rack moves in one direction}}$$

If the pinion reverses:

$$\boxed{\text{Rack moves in the opposite direction}}$$

So:

Clockwise pinion $\rightarrow$ rack upward/downward

Counterclockwise pinion → rack downward/upward

The actual direction depends on which side of the pinion the rack is positioned.

20. Sector rotation

The rack moves the spur gear.

Rack displacement is related to spur rotation by:

$$L = r_{sp} \theta_{sp}$$

Therefore:

$$\theta_{sp} = \frac{L}{r_{sp}}$$

For 50 mm:

$$\begin{aligned}\theta_{sp} &= \frac{50}{22.5} \\ \theta_{sp} &= 2.222 \text{ rad}\end{aligned}$$

Converting to degrees:

$$\boxed{\theta_{sp} = 127.3^\circ}$$

The spur gear then rotates the sector.

Gear ratio:

$$\begin{aligned}i &= \frac{Z_s}{Z_{sp}} \\ i &= \frac{66}{30} \\ \boxed{i} &= 2.2\end{aligned}$$

Therefore:

$$\theta_s = \theta_{sp} \frac{Z_{sp}}{Z_s}$$

For 50 mm rack movement:

$$\begin{aligned}\theta_s &= 127.3 \left(\frac{30}{66} \right) \\ \boxed{\theta_s} &= 57.86^\circ\end{aligned}$$

So 50 mm rack travel produces approximately:

$$\boxed{57.9^\circ}$$

sector rotation.

21. Tooth forces at 20° pressure angle

Normal force:

$$F_n = \frac{F_t}{\cos \phi}$$

For the rack-spur mesh:

$$F_{n,sp} = \frac{297.27}{\cos 20^\circ}$$
$$\boxed{F_{n,sp} = 316.35 \text{ N}}$$

With FOS:

$$\boxed{F_{n,sp,design} = 790.88 \text{ N}}$$

Radial force:

$$F_r = F_t \tan \phi$$
$$F_{r,sp} = 297.27 \tan 20^\circ$$
$$\boxed{F_{r,sp} = 108.18 \text{ N}}$$

For the rack-pinion mesh:

$$F_{n,p} = \frac{247.27}{\cos 20^\circ}$$
$$\boxed{F_{n,p} = 263.14 \text{ N}}$$

and:

$$F_{r,p} = 247.27 \tan 20^\circ$$
$$\boxed{F_{r,p} = 89.98 \text{ N}}$$

22. Complete gear dimensions

Dimension	Pinion	Spur	Sector
Module	1.5 mm	1.5 mm	1.5 mm
Teeth	18	30	66
Pressure angle	20°	20°	20°
Pitch diameter	27 mm	45 mm	99 mm
Pitch radius	13.5 mm	22.5 mm	49.5 mm
Outside diameter	30 mm	48 mm	102 mm
Root diameter	23.25 mm	41.25 mm	95.25 mm
Base diameter	25.37 mm	42.29 mm	93.08 mm
Addendum	1.5 mm	1.5 mm	1.5 mm
Dedendum	1.875 mm	1.875 mm	1.875 mm
Total tooth depth	3.375 mm	3.375 mm	3.375 mm
Circular pitch	4.712 mm	4.712 mm	4.712 mm
Tooth thickness	2.356 mm	2.356 mm	2.356 mm

23. Recommended face width

For your PLA prototype:

$$b = 8 \text{ to } 10 \text{ mm}$$

I recommend:

$$b = 10 \text{ mm}$$

for the pinion and spur.

For the sector:

$$b = 10 \text{ mm}$$

This gives better load distribution and is easier to print robustly.

24. Final proposed design

Pinion:

$$m = 1.5 \text{ mm}$$

$$Z = 18$$

$$d = 27 \text{ mm}$$

$$d_o = 30 \text{ mm}$$

$$b = 10 \text{ mm}$$

Spur:

$$m = 1.5 \text{ mm}$$

$$Z = 30$$

$$d = 45 \text{ mm}$$

$$d_o = 48 \text{ mm}$$

$$b = 10 \text{ mm}$$

Sector:

$$m = 1.5 \text{ mm}$$

$$Z = 66$$

$$d = 99 \text{ mm}$$

$$d_o = 102 \text{ mm}$$

$$b = 10 \text{ mm}$$

Arm:

$$L_a = 300 \text{ mm}$$

Rack:

$$m = 1.5 \text{ mm}$$

$$p = 4.712 \text{ mm}$$

$$12 \text{ teeth minimum}$$

$$\text{Toothed length} = 56.55 \text{ mm}$$

$$\text{Recommended physical length} \approx 75 \text{ to } 80 \text{ mm}$$

25. Complete force and torque design values

Parameter	Actual	Design, SF = 2.5
Payload force	49.05 N	122.63 N
Sector torque	14.715 N·m	36.79 N·m
Sector tangential force	297.27 N	743.18 N
Direct rack force	50 N	125 N
Net rack force	247.27 N	618.18 N
Spur torque	6.688 N·m	16.72 N·m
Pinion torque	3.338 N·m	8.35 N·m
Rack-spur normal force	316.35 N	790.88 N
Rack-pinion normal force	263.14 N	657.86 N

26. One important design issue

Your 300 N value is not an additional 300 N that should be added to the 50 N force.

The correct static equilibrium is:

$$F_{net} = 297.27 - 50 = 247.27 \text{ N}$$

But the gear teeth transmitting the payload load must still be checked against approximately:

$$297.27 \text{ N actual}$$

and:

$$743.18 \text{ N design}$$

This distinction is important.

Also, because this is PLA and the design load is high, I would not finalize the mechanism until we perform the Lewis bending stress calculation for the pinion, spur, sector and rack. That calculation needs to use the 10 mm face width and the actual PLA allowable stress, including a suitable reduction for 3D-printing layer orientation.